Short Guide: Curating VMR Projects for ML Training

1 Overview

This guide describes how to organize and prepare Vascular Model Repository (VMR) projects for machine learning training. The goal is to create a clean, consistent dataset where each patient or project has a matched image volume, surface model, and centerline file.

2 Required Dataset Structure

Each curated patient or project should be organized into the following structure:

```
data_dir/
|- images/
|- surfaces/
|- centerlines/
```

Each folder should contain one file per patient or project, with matching file names across folders whenever possible.

Example:

```
data_dir/
|- images/
|  |- patient_001.nii.gz
|- surfaces/
|  |- patient_001.vtp
|- centerlines/
   |- patient_001.vtp
```

Accepted image formats may include:

```
.nii.gz
.vti
.mha
```

Surface models and centerlines should generally be stored as:

```
.vtp
```

The base file name should match across all three folders so the training pipeline can associate each image with its corresponding surface model and centerline.

3 Image Files

For each VMR project, locate the primary medical image volume used for segmentation or modeling.

Place the image file in:

`data_dir/images/`

Before adding the file, confirm that:

- The image is a valid 3D volume.
- The file opens correctly in SimVascular, 3D Slicer, ParaView, or another viewer.
- The file name matches the corresponding surface and centerline files.
- The image is not duplicated or from an unrelated project.

4 Surface Models

Surface models should be placed in:

`data_dir/surfaces/`

Surface models are typically `.vtp` files representing the vessel geometry.

Some VMR projects may not have an obvious surface model in the main project directory. If a surface model is missing, check the project's `meshes/` folder. In some cases, usable surface geometry can be found there and copied or exported as the patient's surface model.

Before adding the surface file, confirm that:

- The surface opens correctly.
- The vessel geometry is complete.
- The file corresponds to the correct patient or project.
- The model is not an intermediate, broken, or unrelated mesh file.

5 Centerline Files

Centerline files should be placed in:

`data_dir/centerlines/`

Some VMR projects may already include centerlines. If centerlines are missing, they should be extracted from the surface model.

5.1 Centerline Extraction in SimVascular

Centerline extraction can be performed in SimVascular using the surface model. In SimVascular, open the project and go to the **Modeling** window. Within the Modeling tools, use the **Centerline Extraction** tab.

To extract centerlines, the model must have defined faces. These faces identify which parts of the model are vessel walls and which parts are open vessel caps. During centerline extraction, one cap is selected as the **source** face, and all other relevant caps are selected as **target** faces. The source face is usually chosen at the main inlet or proximal vessel opening. The target faces are typically the outlet caps or distal branch openings.

If the surface model does not already have faces defined, this must be done before centerline extraction. The model should be separated into wall and cap faces so that SimVascular can identify the vessel inlet, outlets, and wall surface. In general:

- The **wall** face should correspond to the vessel wall surface.
- Each **cap** should correspond to an open end of the vessel model.
- One cap should be chosen as the **source**.
- All other outlet caps should be chosen as **targets**.

After the source and target caps are selected, run the centerline extraction tool. Review the generated centerlines visually to make sure they follow the vessel lumen and extend from the source cap to the intended target caps. If the centerline is incomplete, incorrect, or follows the wrong path, check that the correct caps were selected and that the model faces are properly defined.

After extraction, export the centerline as a `.vtp` file and save it with the same base name as the image and surface model.

Example:

```
images/patient_001.nii.gz
surfaces/patient_001.vtp
centerlines/patient_001.vtp
```

Before adding the centerline file, confirm that:

- The centerline follows the vessel path correctly.
- The centerline begins at the selected source cap.
- The centerline reaches the intended target caps.
- Major branches are represented.
- The centerline belongs to the correct surface model.
- The file opens correctly in a viewer.

6 Naming Convention

Use a consistent naming convention for all curated files. The safest format is:

```
project_or_patient_id.ext
```

Example:

```
OSMSC0001.nii.gz
OSMSC0001.vtp
OSMSC0001.vtp
```

Avoid spaces, special characters, or inconsistent labels.

Good examples:

```
patient_001.nii.gz
patient_001.vtp
```

Avoid:

```
Patient 1 final image.nii.gz
surface-new-copy.vtp
centerline_test.vtp
```

7 Quality Control Checklist

Before marking a project as ready for ML training, confirm that:

- The image file exists.
- The surface model exists.
- The centerline file exists or has been extracted.
- All three files use the same base name.
- Files open successfully in a visualization tool.
- The surface and centerline match the image anatomy.
- There are no duplicate, temporary, or unrelated files.
- Missing surfaces were checked for under the `meshes/` folder.
- Missing centerlines were extracted from the surface model when needed.

8 Final Curated Dataset Example

```
data_dir/
|- images/
|  |- patient_001.nii.gz
|  |- patient_002.mha
|  |- patient_003.vti
|- surfaces/
|  |- patient_001.vtp
|  |- patient_002.vtp
|  |- patient_003.vtp
|- centerlines/
   |- patient_001.vtp
   |- patient_002.vtp
   |- patient_003.vtp
```

A project should only be included in the final ML training dataset once the image, surface, and centerline files are all present, correctly matched, and visually checked.